

Inverting the Subfactorial at Infinity

Bell-Sector Transseries, Inverse-Gamma Geometry,
and a General Reversion Calculus for Rapid Cores with Tiny Oscillatory Tails

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Abstract

The integer subfactorial $D_n = !n$ is asymptotic to $n!/e$, but its exponentially small absolute correction has a complete inverse-power expansion whose coefficients are Bell numbers. Inverting this relation is subtler than inverting Stirling's formula: a noninteger subfactorial requires a choice of analytic continuation, the dominant inverse is the principal inverse gamma function, and the Bell correction creates a new family of sectors beyond the ordinary Lambert–Stirling scale.

We begin with the lateral analytic continuations

$$\mathfrak{D}_\sigma(z) = \frac{\Gamma_\sigma(z+1, -1)}{e}, \quad \sigma \in \{+1, -1\},$$

where the negative argument is reached with argument $\sigma\pi$. For $\Re z > -1$ they admit the exact decomposition

$$\mathfrak{D}_\sigma(z) = \frac{\Gamma(z+1)}{e} + e^{i\sigma\pi z} \mathcal{J}(z), \quad \mathcal{J}(z) = \frac{1}{e} \int_0^1 e^{tz} dt = \frac{1}{e} \sum_{j \geq 0} \frac{1}{j!(z+j+1)}.$$

The endpoint integral satisfies

$$\mathcal{J}(x) \sim \sum_{k \geq 1} (-1)^{k-1} \frac{B_k}{x^k},$$

and we give an exact signed remainder whose absolute value is bounded by $B_{m+1}x^{-m-1}$ after m terms. Thus the Bell expansion is not merely an integer identity; it is the slow coefficient expansion of an analytic oscillatory tail.

Let $\mathfrak{a}(y)$ be the large real solution of $\Gamma(\mathfrak{a}(y)+1)/e = y$, and put $P(x) = \psi(x+1)$. There is, for each large positive y , a unique inverse branch $X_\sigma(y)$ of $\mathfrak{D}_\sigma$ satisfying $X_\sigma(y) - \mathfrak{a}(y) \rightarrow 0$. Its complete Bell-sector transseries is

$$X_\sigma(y) \sim \mathfrak{a}(y) + \sum_{n \geq 1} \frac{e^{i\sigma n \pi \mathfrak{a}(y)}}{y^n} C_{n,\sigma}(\mathfrak{a}(y)),$$

where

$$C_{n,\sigma}(x) = \frac{(-1)^n}{n!} (\mathcal{L}_{n-1,n}^{(\sigma)} \circ \cdots \circ \mathcal{L}_{1,n}^{(\sigma)}) \left(\frac{\mathcal{J}(x)^n}{P(x)} \right), \quad \mathcal{L}_{r,n}^{(\sigma)} f = \frac{f'}{P} + \left(\frac{i\sigma n \pi}{P} - r \right) f.$$

We calculate the first sectors explicitly, expand their coefficients in Bell numbers and logarithms, prove fixed-sector error estimates, and verify the formula numerically. The core $\mathfrak{a}$ is itself expanded to arbitrary order by a centered Stirling inversion. If $\lambda = \log(e y / \sqrt{2\pi})$, $w = W_0(\lambda/e)$, and $u_0 = \lambda/w$, then

$$\mathfrak{a}(y) = -\frac{1}{2} + u_0 + \sum_{m \geq 1} \frac{p_m(w)}{u_0^{2m-1}},$$

with an explicit triangular recurrence for the rational functions p_m .

The final part abstracts the calculation. For a rapidly varying core $\mathcal{A}(x)$ and a small slow or oscillatory tail $\mathcal{R}(x)$, we prove the all-orders identity

$$(\mathcal{A} + \mathcal{R})^{-1}(y) \sim a(y) + \sum_{n \geq 1} \frac{(-1)^n}{n!} \frac{d^{n-1}}{dy^{n-1}} [a'(y) \mathcal{R}(a(y))^n], \quad a = \mathcal{A}^{-1},$$

and convert it into a valuation-raising operator calculus. This supplies a reusable transseries inversion theory for factorial, gamma-type, Barnes-type, and endpoint-Mellin perturbations. We also treat a real median interpolation and the genuinely discrete generalized inverse, whose staircase nature must be kept separate from any smooth inverse transseries.

Main result

Let

$$\mathcal{A}(x) = \frac{\Gamma(x+1)}{e}, \quad \mathcal{J}(x) = \frac{1}{e} \int_0^1 e^t t^x dt, \quad \mathcal{A}(\mathfrak{a}(y)) = y, \quad P(x) = \psi(x+1).$$

For $y \rightarrow +\infty$, the two lateral inverse branches have the sector-exact normal form

$$\begin{aligned} X_\sigma(y) = \mathfrak{a}(y) &- \frac{e^{i\sigma\pi\mathfrak{a}(y)}}{y} \frac{\mathcal{J}(\mathfrak{a}(y))}{P(\mathfrak{a}(y))} \\ &+ \frac{e^{2i\sigma\pi\mathfrak{a}(y)}}{y^2} \left[\frac{\mathcal{J}\mathcal{J}'}{P^2} + \frac{i\sigma\pi\mathcal{J}^2}{P^2} - \frac{\mathcal{J}^2 P'}{2P^3} - \frac{\mathcal{J}^2}{2P} \right]_{x=\mathfrak{a}(y)} \\ &+ \mathcal{O}\left(\frac{1}{y^3 \mathfrak{a}(y)^3 \log \mathfrak{a}(y)}\right). \end{aligned}$$

Writing $a = \mathfrak{a}(y)$ and $\ell = \log a$, the first sector begins as

$$-\frac{e^{i\sigma\pi a}}{y} \left[\frac{1}{a\ell} - \frac{4\ell+1}{2a^2\ell^2} + \frac{60\ell^2+13\ell+3}{12a^3\ell^3} - \frac{360\ell^3+64\ell^2+14\ell+3}{24a^4\ell^4} + \mathcal{O}\left(\frac{1}{a^5\ell}\right) \right]$$

and the second begins as

$$\frac{e^{2i\sigma\pi a}}{y^2} \left[-\frac{1}{2a^2\ell} + \frac{i\sigma\pi}{a^2\ell^2} + \frac{8\ell^2-3\ell-2-16i\sigma\pi\ell-4i\sigma\pi}{4a^3\ell^3} + \mathcal{O}\left(\frac{1}{a^4\ell}\right) \right].$$

At general sector number n ,

$$C_{n,\sigma}(a) \sim -\frac{1}{n} \frac{\mathcal{J}(a)^n}{P(a)} \sim -\frac{1}{na^n \log a}.$$

The core is

$$\begin{aligned} \mathfrak{a}(y) = & -\frac{1}{2} + u_0 + \frac{1}{24(1+w)u_0} - \frac{14w^2+38w+29}{5760(1+w)^3u_0^3} \\ & + \frac{2232w^4+10104w^3+17634w^2+14234w+4577}{2903040(1+w)^5u_0^5} + \mathcal{O}(u_0^{-7}), \end{aligned}$$

where $\lambda = \log(ey/\sqrt{2\pi})$, $w = W_0(\lambda/e)$, and $u_0 = \lambda/w$.

A branch issue that cannot be ignored

The integer sequence D_n has no ordinary inverse function. Moreover, the standard incomplete-gamma continuation is complex-valued between the integers. Consequently, three distinct objects must not be conflated: (i) a lateral complex inverse branch X_σ ; (ii) a chosen real interpolation and its inverse; and (iii) the integer-valued generalized inverse $\min\{n : D_n \geq y\}$. All three are analyzed below, but they have different asymptotic meanings.

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1 Introduction and orientation

For an integer $n \geq 0$, the subfactorial or derangement number

$$D_n = !n = n! \sum_{j=0}^n \frac{(-1)^j}{j!} \quad (1.1)$$

counts permutations of n objects having no fixed point. Its leading asymptotic behavior is the elementary relation $D_n \sim n!/e$. A much more informative formula, emphasized by Hassani [8], is

$$D_n = \frac{n!}{e} + \sum_{k=1}^m (-1)^{n+k-1} \frac{B_k}{n^k} + \mathcal{O}(n^{-m-1}), \quad (1.2)$$

where m is fixed, B_k is the k th Bell number, and the constant in the remainder can be taken no larger than B_{m+1} .

The purpose of this article is to reverse (1.2). The naive instruction “solve $D_x = y$ ” hides four layers of structure.

1. The integer sequence has to be embedded in a continuous or analytic object before it has a local inverse in the usual sense.
2. Its dominant part is a gamma function, so the first approximation is an inverse gamma function rather than merely a logarithm.
3. Inverting Stirling’s formula introduces a Lambert W function and a polynomial–logarithmic expansion.
4. The Bell correction is absolutely tiny compared with $\Gamma(x+1)$, but after inversion it survives as a new hierarchy of sectors of order y^{-n} , each carrying an oscillatory phase and its own inverse-power/logarithmic coefficient expansion.

The linked ProveIt draft collection develops arithmetic, composition, and reversion in polynomial–logarithmic transseries and analyzes related near-identity inversions [12]. The present problem adds a different geometry: a *rapid core* whose logarithmic derivative grows slowly, plus a bounded or algebraically decaying correction. The correct small parameter is not $1/x$ alone. It is the relative tail $\mathcal{R}(x)/\mathcal{A}(x)$, which becomes a power of $1/y$ after the core has been inverted.

1.1 The three scales

Set $Y = \log y$. The large inverse-gamma variable satisfies roughly

$$\mathfrak{a}(y) \asymp \frac{Y}{\log Y}.$$

Its asymptotic description contains the following nested scales.

Lambert scale.

The leading object $Y/W(Y/e)$ resums the repeated logarithms generated by reversing $x \log x - x$.

Stirling scale.

Centered Stirling corrections form an odd inverse-power series in $u_0^{-1} \asymp (\log Y)/Y$, with rational functions of $W(Y/e)$ as coefficients.

Bell sectors.

The subfactorial tail produces sectors proportional to $y^{-n} = e^{-nY}$. Inside the n th sector are powers of $\mathfrak{a}(y)^{-1}$, $\log \mathfrak{a}(y)^{-1}$, and phases $\exp(in\pi\mathfrak{a}(y))$.

Thus a typical monomial in the final expansion has the shape

$$e^{-nY} e^{in\sigma\pi\mathfrak{a}(e^Y)} \mathfrak{a}(e^Y)^{-p} (\log \mathfrak{a}(e^Y))^{-q}, \quad n, p, q \geq 0. \quad (1.3)$$

The outer index n is the decisive valuation. At each fixed n , the remaining coefficient admits an ordinary slow asymptotic expansion.

1.2 Why the core is kept exact for as long as possible

The inverse gamma function has its own divergent Stirling expansion. One can refine it hyperasymptotically through Binet-type remainder formulae and exponentially improved gamma expansions [5, 9]. Those refinements need not be mixed prematurely with the Bell sectors. Our master formula treats $\mathfrak{a}(y)$ exactly and expands only the perturbation around it. This “sector-exact normal form” has three advantages:

- it is independent of how the inverse gamma core is evaluated;
- it preserves the rigorous ordering by powers of y^{-1} ;
- any preferred algebraic, convergent, numerical, or hyperasymptotic representation of $\mathfrak{a}$ can be substituted afterward.

1.3 Main contributions

The main results may be summarized as follows.

1. We derive an exact decomposition of every lateral incomplete-gamma continuation into a gamma core and an endpoint-Mellin tail.
2. We prove the Bell-number expansion with an explicit signed remainder, including derivative expansions needed by reversion.
3. We obtain a centered Lambert–Stirling expansion of the large inverse gamma branch to arbitrary order, with a triangular coefficient recurrence and explicit initial terms.
4. We prove an all-orders Lagrange–Bürmann formula for inversion around a rapid core and rewrite it as a valuation-raising differential operator calculus.
5. We specialize that calculus to the principal and conjugate subfactorial continuations, obtaining explicit y^{-1} and y^{-2} sectors and a closed coefficient formula for every sector.
6. We analyze the real median interpolation and the discrete generalized inverse separately, including exponentially small threshold shifts.
7. We generalize the Bell mechanism to arbitrary endpoint-Mellin tails and explain which parts of the inverse transseries are universal.

2 Subfactorials beyond the integers

2.1 Elementary integer formulae

The inclusion–exclusion formula (1.1) gives the familiar recurrences

$$D_n = nD_{n-1} + (-1)^n = (n-1)(D_{n-1} + D_{n-2}), \quad D_0 = 1, \quad D_1 = 0. \quad (2.1)$$

The first form already hints at a factorial core plus a parity-dependent small correction. To expose that correction exactly, write

$$e^{-1} - \sum_{j=0}^n \frac{(-1)^j}{j!} = \sum_{j=n+1}^{\infty} \frac{(-1)^j}{j!}. \quad (2.2)$$

Repeated integration by parts or Taylor’s theorem with integral remainder will convert this tail into the endpoint integral introduced below.

2.2 Lateral incomplete-gamma continuations

The upper incomplete gamma function at a negative argument depends on the chosen determination of the logarithm in t^z . We use the following notation.

Definition 2.1 (Lateral subfactorials). For $\sigma \in \{+1, -1\}$, let

$$\mathfrak{D}_\sigma(z) = \frac{1}{e} \Gamma_\sigma(z+1, -1), \quad (2.3)$$

where the integration path reaches the negative real axis with $\arg(-1) = \sigma\pi$. The choice $\sigma = +1$ is the usual principal boundary value. More generally, after winding $k \in \mathbb{Z}$ times around the origin one obtains a continuation with phase $\omega_k = (2k+1)\pi$.

Define the two building blocks

$$\mathcal{A}(z) = \frac{\Gamma(z+1)}{e}, \quad \mathcal{J}(z) = \frac{1}{e} \int_0^1 e^{t^z} dt, \quad \Re z > -1. \quad (2.4)$$

Theorem 2.2 (Exact core–tail decomposition). For $\Re z > -1$,

$$\mathfrak{D}_\sigma(z) = \mathcal{A}(z) + e^{i\sigma\pi z} \mathcal{J}(z). \quad (2.5)$$

More generally, the branch with $\arg(-1) = (2k+1)\pi$ is

$$\mathfrak{D}_k(z) = \mathcal{A}(z) + e^{i(2k+1)\pi z} \mathcal{J}(z). \quad (2.6)$$

Every one of these functions continues to an entire function of z , and all of them agree with D_n at nonnegative integers.

Proof. For $s = z+1$ and the upper lateral path, parameterize the segment from 0 to -1 by $t = -u$ with $\arg t = \pi$. Then

$$\gamma_+(s, -1) = \int_0^{-1} t^{s-1} e^{-t} dt = e^{i\pi s} \int_0^1 u^{s-1} e^u du.$$

The relation $\Gamma_+(s, -1) = \Gamma(s) - \gamma_+(s, -1)$ and $e^{i\pi(z+1)} = -e^{i\pi z}$ yield

$$\Gamma_+(z+1, -1) = \Gamma(z+1) + e^{i\pi z} \int_0^1 u^z e^u du.$$

The lower path is identical with π replaced by $-\pi$, and additional windings replace it by $(2k+1)\pi$.

For fixed nonzero second argument, the upper incomplete gamma function is entire in its first parameter. One can also see cancellation directly from (3.1) below: at $z = -m$, $m \geq 1$, the residues of $\mathcal{A}(z)$ and $e^{i(2k+1)\pi z} \mathcal{J}(z)$ are respectively $(-1)^{m-1}/(e(m-1)!)$ and $(-1)^m/(e(m-1)!)$. They cancel. Finally, at $z = n \in \mathbb{N}_0$, the phase is $(-1)^n$; comparison with the integral remainder for the exponential series gives $\mathfrak{D}_k(n) = D_n$. $\square$

Corollary 2.3 (Exact integer correction). *For every $n \geq 0$,*

$$D_n = \frac{n!}{e} + (-1)^n \mathcal{J}(n). \quad (2.7)$$

Moreover,

$$0 < \mathcal{J}(x) < \frac{1}{x+1}, \quad x > -1. \quad (2.8)$$

Consequently, D_n is the nearest integer to $n!/e$ for every $n \geq 1$.

Proof. The first assertion is (2.5) at an integer. Since $0 < e^{t-1} < 1$ for $0 \leq t < 1$,

$$0 < \mathcal{J}(x) = \int_0^1 e^{t-1} t^x dt < \int_0^1 t^x dt = \frac{1}{x+1}.$$

For $n \geq 1$ this is strictly less than $1/2$, proving the nearest-integer statement. $\square$

2.3 A canonical real interpolation

The lateral values are conjugate on the real axis. Their arithmetic mean is therefore a natural real entire interpolation.

Definition 2.4 (Median subfactorial). Define

$$\mathfrak{D}_{\text{med}}(z) = \frac{\mathfrak{D}_+(z) + \mathfrak{D}_-(z)}{2} = \mathcal{A}(z) + \cos(\pi z) \mathcal{J}(z). \quad (2.9)$$

The residue cancellation in the proof of Theorem 2.2 shows that $\mathfrak{D}_{\text{med}}$ is entire. It is real-valued on the real axis and agrees with D_n at every nonnegative integer. It is not the only possible real interpolation; without a normalization condition, infinitely many entire functions can be added while vanishing on the integers. It is, however, the canonical median of the two nearest lateral determinations, and its inverse will provide a useful real counterpart to the principal complex inverse.

Proposition 2.5 (Eventual monotonicity). *The real function $\mathfrak{D}_{\text{med}}(x)$ is strictly increasing for all sufficiently large x . The same is true of the modulus-dominant gamma core $\mathcal{A}(x)$, and*

$$\mathfrak{D}'_{\text{med}}(x) = \mathcal{A}(x) \psi(x+1) + \mathcal{J}'(x) \cos(\pi x) - \pi \mathcal{J}(x) \sin(\pi x). \quad (2.10)$$

Proof. As $x \rightarrow \infty$, $\mathcal{A}(x) \rightarrow \infty$ and $\psi(x+1) \sim \log x$. On the other hand, $\mathcal{J}(x) = \mathcal{O}(x^{-1})$ and $\mathcal{J}'(x) = \mathcal{O}(x^{-2})$, as follows either from the integral or from Theorem 3.3. Hence the first term in (2.10) dominates the other two and is positive. $\square$

3 The endpoint tail and its Bell-number expansion

The small function $\mathcal{J}$ contains the entire derangement correction. Its asymptotics are especially transparent because it is simultaneously an endpoint integral, a meromorphic partial-fraction series, and a Poisson moment transform.

3.1 Exact series and meromorphic continuation

Proposition 3.1 (Partial-fraction representation). *For $\Re z > -1$,*

$$\mathcal{J}(z) = \frac{1}{e} \sum_{j=0}^{\infty} \frac{1}{j!(z+j+1)}. \quad (3.1)$$

The series converges locally uniformly away from $z = -1, -2, \dots$ and gives the meromorphic continuation of $\mathcal{J}$ to $\mathbb{C}$, with simple residues

$$\text{Res}_{z=-m} \mathcal{J}(z) = \frac{1}{e(m-1)!}, \quad m \geq 1. \quad (3.2)$$

For every nonnegative integer r ,

$$\mathcal{J}^{(r)}(z) = \frac{(-1)^r r!}{e} \sum_{j=0}^{\infty} \frac{1}{j!(z+j+1)^{r+1}}. \quad (3.3)$$

Proof. Expand $e^t = \sum_{j \geq 0} t^j / j!$ in (2.4) and integrate termwise. Uniform convergence on $[0, 1]$ and local uniform convergence in z on compact subsets of $\Re z > -1$ justify the interchange. The resulting series converges normally on compact subsets avoiding its displayed poles, which proves the continuation and residues. Termwise differentiation gives (3.3). $\square$

3.2 Dobiński moments shifted by one

Recall Dobiński's formula

$$B_k = \frac{1}{e} \sum_{j=0}^{\infty} \frac{j^k}{j!}, \quad k \geq 0. \quad (3.4)$$

A shifted version is the form naturally required by (3.1).

Lemma 3.2 (Shifted Poisson moment). *For every integer $r \geq 0$,*

$$\frac{1}{e} \sum_{j=0}^{\infty} \frac{(j+1)^r}{j!} = B_{r+1}. \quad (3.5)$$

Proof. Let N be Poisson with mean 1. Then the left-hand side is $\mathbb{E}(N+1)^r$. Expanding by the binomial theorem and using $\mathbb{E}N^q = B_q$ gives

$$\mathbb{E}(N+1)^r = \sum_{q=0}^r \binom{r}{q} B_q.$$

The Bell recurrence $B_{r+1} = \sum_{q=0}^r \binom{r}{q} B_q$ completes the proof. Equivalently, one may compare coefficients in $e^{e^t-1+t} = \frac{d}{dt} e^{e^t-1}$. $\square$

3.3 The Bell expansion with an exact remainder

Theorem 3.3 (Bell expansion and sharp elementary bound). *Let $m \geq 1$ be fixed and $x > 0$. Then*

$$\mathcal{J}(x) = \sum_{k=1}^m (-1)^{k-1} \frac{B_k}{x^k} + R_m(x), \quad (3.6)$$

where the remainder has the exact representation

$$R_m(x) = \frac{(-1)^m}{e x^m} \sum_{j=0}^{\infty} \frac{(j+1)^m}{j!(x+j+1)}. \quad (3.7)$$

In particular,

$$(-1)^m R_m(x) > 0, \quad |R_m(x)| \leq \frac{B_{m+1}}{x^{m+1}}. \quad (3.8)$$

Consequently,

$$\mathcal{J}(x) \sim \frac{1}{x} - \frac{2}{x^2} + \frac{5}{x^3} - \frac{15}{x^4} + \frac{52}{x^5} - \frac{203}{x^6} + \frac{877}{x^7} - \frac{4140}{x^8} + \cdots. \quad (3.9)$$

Proof. For $a > 0$ the finite geometric identity gives

$$\frac{1}{x+a} = \sum_{k=1}^m (-1)^{k-1} \frac{a^{k-1}}{x^k} + (-1)^m \frac{a^m}{x^m(x+a)}. \quad (3.10)$$

Insert $a = j+1$ into (3.1) and sum over j . By Theorem 3.2, the coefficient of x^{-k} is $(-1)^{k-1} B_k$, and the remaining term is (3.7). Every summand in the latter is positive after multiplication by $(-1)^m$. Since $(x+j+1)^{-1} \leq x^{-1}$,

$$|R_m(x)| \leq \frac{1}{e x^{m+1}} \sum_{j=0}^{\infty} \frac{(j+1)^m}{j!} = \frac{B_{m+1}}{x^{m+1}}.$$

The displayed coefficients are the first Bell numbers. □

Corollary 3.4 (Hassani's derangement expansion). *For every fixed $m \geq 1$ and every integer $n \geq 1$,*

$$D_n = \frac{n!}{e} + \sum_{k=1}^m (-1)^{n+k-1} \frac{B_k}{n^k} + (-1)^n R_m(n), \quad (3.11)$$

where $|R_m(n)| \leq B_{m+1} n^{-m-1}$.

Proof. Substitute (3.6) into (2.7). □

The proof shows slightly more than a Poincaré expansion: successive truncations alternately bound $\mathcal{J}(x)$. This sign information is useful for certified approximations.

3.4 Derivative expansions

The inverse coefficients involve derivatives of $\mathcal{J}$. They can be obtained by differentiating the asymptotic expansion, but it is useful to record the result explicitly.

Proposition 3.5 (Differentiated Bell expansion). *For fixed integers $r \geq 0$ and $m \geq 1$,*

$$\mathcal{J}^{(r)}(x) = \sum_{k=1}^m (-1)^{r+k-1} (k)_r B_k x^{-k-r} + \mathcal{O}_{m,r}(x^{-m-r-1}), \quad (3.12)$$

where $(k)_r = k(k+1) \cdots (k+r-1)$ is the rising factorial. In particular,

$$\mathcal{J}'(x) = -x^{-2} + 4x^{-3} - 15x^{-4} + 60x^{-5} - 260x^{-6} + \cdots, \quad (3.13)$$

$$\mathcal{J}''(x) = 2x^{-3} - 12x^{-4} + 60x^{-5} - 300x^{-6} + 1560x^{-7} + \cdots. \quad (3.14)$$

Proof. Apply (3.3) and expand $(x+j+1)^{-r-1}$ in powers of x^{-1} , or differentiate (3.6) with a remainder obtained from (3.7). For fixed r, m , the differentiated remainder is bounded by a convergent Poisson moment and has the asserted order. □

3.5 Watson's lemma and the origin of the Bell numbers

There is a conceptual explanation that generalizes immediately. With $t = e^{-s}$,

$$\mathcal{J}(x) = \int_0^\infty e^{-(x+1)s} \exp(e^{-s} - 1) ds. \quad (3.15)$$

The Bell exponential generating function gives

$$\exp(e^{-s} - 1) = \sum_{r=0}^{\infty} B_r \frac{(-s)^r}{r!}. \quad (3.16)$$

Watson's lemma therefore yields the equivalent shifted expansion

$$\mathcal{J}(x) \sim \sum_{r=0}^{\infty} \frac{(-1)^r B_r}{(x+1)^{r+1}}. \quad (3.17)$$

Re-expanding $(x+1)^{-r-1}$ in x^{-1} converts (3.17) into (3.9). The shift by one in Theorem 3.2 is exactly the algebraic shadow of the Jacobian $dt = -e^{-s} ds$.

Structural summary

The Bell numbers enter before inversion. They are the endpoint derivatives of $\exp(e^{-s} - 1)$, equivalently the shifted Poisson moments of the partial fractions in (3.1). In the inverse problem they remain inside the slowly varying coefficient functions of the y^{-n} sectors; inversion does not replace them by a different combinatorial sequence, but combines them through products, derivatives, and Bell polynomials of a second kind.

4 The inverse-gamma core

Before the Bell perturbation can be inverted, the factorial core must be understood. Let $x_* > 0$ denote the unique positive zero of $\psi(x+1)$; then $\mathcal{A}$ is strictly increasing on (x_*, ∞) . For every sufficiently large y there is a unique real number $\mathfrak{a}(y) > x_*$ satisfying

$$\mathcal{A}(\mathfrak{a}(y)) = \frac{\Gamma(\mathfrak{a}(y) + 1)}{e} = y. \quad (4.1)$$

In terms of the principal real inverse gamma branch,

$$\mathfrak{a}(y) = \Gamma_0^{-1}(ey) - 1. \quad (4.2)$$

The theory of inverse gamma branches has been studied analytically and computationally by Uchiyama, Pedersen, Borwein, Corless, Jeffrey, and their collaborators [13, 11, 1, 3].

4.1 Exact differential identities

Set

$$P(x) = \psi(x+1) = \frac{\mathcal{A}'(x)}{\mathcal{A}(x)}. \quad (4.3)$$

Implicit differentiation of (4.1) gives

$$\mathfrak{a}'(y) = \frac{1}{yP(\mathfrak{a}(y))}. \quad (4.4)$$

Further derivatives are obtained recursively. For example, writing $a = \mathfrak{a}(y)$,

$$\mathfrak{a}''(y) = -\frac{1}{y^2 P(a)} - \frac{P'(a)}{y^2 P(a)^3}, \quad (4.5)$$

$$\mathfrak{a}'''(y) = \frac{2}{y^3 P(a)} + \frac{3P'(a)}{y^3 P(a)^3} - \frac{P''(a)}{y^3 P(a)^4} + \frac{3P'(a)^2}{y^3 P(a)^5}. \quad (4.6)$$

These identities are useful both for error propagation and for direct Lagrange–Bürmann calculations.

4.2 Centered Stirling coordinates

The ordinary Stirling expansion becomes particularly sparse after the half-unit shift

$$u = a + \frac{1}{2}. \quad (4.7)$$

For $u \rightarrow +\infty$,

$$\log \Gamma\left(u + \frac{1}{2}\right) - \frac{1}{2} \log(2\pi) \sim u \log u - u + \sum_{m=1}^{\infty} \frac{c_m}{u^{2m-1}}, \quad (4.8)$$

where

$$c_m = \frac{B_{2m}(1/2)}{2m(2m-1)} = \frac{(2^{1-2m} - 1)B_{2m}}{2m(2m-1)}. \quad (4.9)$$

The first coefficients are

$$c_1 = -\frac{1}{24}, \quad c_2 = \frac{7}{2880}, \quad c_3 = -\frac{31}{40320}, \quad c_4 = \frac{127}{215040}. \quad (4.10)$$

Since $\Gamma(a+1) = ey$, define

$$\lambda = \log \frac{ey}{\sqrt{2\pi}}. \quad (4.11)$$

Then the exact inversion problem has the asymptotic normal form

$$u \log u - u + \sum_{m \geq 1} c_m u^{-(2m-1)} \sim \lambda. \quad (4.12)$$

4.3 The Lambert seed

Ignore the inverse-power corrections in (4.12). The remaining equation

$$u_0(\log u_0 - 1) = \lambda \quad (4.13)$$

can be solved exactly by Lambert W . Put

$$w = W_0(\lambda/e), \quad u_0 = \frac{\lambda}{w} = e^{w+1}. \quad (4.14)$$

Then $\log u_0 - 1 = w$, and (4.13) follows. This is the natural zeroth approximation to the principal inverse gamma branch; it is the centered form of the inverse-factorial approximation discussed by Borwein and Corless [1].

4.4 All-orders centered inversion

Theorem 4.1 (Centered inverse-gamma expansion). *As $y \rightarrow +\infty$, with λ, w, u_0 defined by (4.11)–(4.14),*

$$\mathfrak{a}(y) = -\frac{1}{2} + u_0 + \sum_{m=1}^M \frac{p_m(w)}{u_0^{2m-1}} + \mathcal{O}_M(u_0^{-2M-1}), \quad (4.15)$$

where each $p_m(w)$ is a rational function regular for $w \geq 0$. The first four are

$$p_1(w) = \frac{1}{24(1+w)}, \quad (4.16)$$

$$p_2(w) = -\frac{14w^2 + 38w + 29}{5760(1+w)^3}, \quad (4.17)$$

$$p_3(w) = \frac{2232w^4 + 10104w^3 + 17634w^2 + 14234w + 4577}{2903040(1+w)^5}, \quad (4.18)$$

$$p_4(w) = -\frac{1}{1393459200(1+w)^7} \left(822960w^6 + 5230296w^5 + 13944036w^4 + 20000424w^3 + 16328586w^2 + 7230574w + 1368337 \right). \quad (4.19)$$

The coefficients are determined uniquely by the triangular recurrence in (4.23) below.

Proof. Write $r = u_0^{-1}$ and seek

$$u = u_0 + \Delta(r), \quad \Delta(r) = \sum_{m \geq 1} p_m(w) r^{2m-1}. \quad (4.20)$$

Since $\log u_0 - 1 = w$, subtraction of $u_0 \log u_0 - u_0 = \lambda$ from (4.12) gives

$$0 \sim \mathcal{E}(\Delta; r, w) := w\Delta + (r^{-1} + \Delta) \log(1 + r\Delta) + \sum_{j \geq 1} c_j r^{2j-1} (1 + r\Delta)^{-(2j-1)}. \quad (4.21)$$

The coefficient of r^{2m-1} is affine in p_m , and its coefficient of p_m is $1 + w$. Therefore the equations can be solved successively. The absence of even powers follows inductively from the odd centered Stirling series and the fact that $r\Delta$ is even. Direct coefficient extraction gives (4.16)–(4.19). Standard asymptotic implicit-function arguments, or substitution followed by one Newton correction, prove the stated remainder at each fixed M . $\square$

For an explicit recurrence, let

$$\Delta_{<m}(r) = \sum_{j=1}^{m-1} p_j(w) r^{2j-1}. \quad (4.22)$$

Then

$$p_m(w) = -\frac{1}{1+w} [r^{2m-1}] \left\{ w\Delta_{<m} + (r^{-1} + \Delta_{<m}) \log(1 + r\Delta_{<m}) + \sum_{j=1}^m c_j r^{2j-1} (1 + r\Delta_{<m})^{-(2j-1)} \right\}. \quad (4.23)$$

This recurrence uses only addition, multiplication, rational powers of a unit series, and coefficient extraction. It is therefore directly implementable in any formal power-series system.

4.5 Expansion into iterated logarithms

The Lambert seed can itself be expanded. Let

$$L_1 = \log(\lambda/e), \quad L_2 = \log L_1. \quad (4.24)$$

The standard large- z expansion of $W_0(z)$ [4] implies

$$\begin{aligned} \frac{1}{W_0(\lambda/e)} &= \frac{1}{L_1} + \frac{L_2}{L_1^2} + \frac{L_2^2 - L_2}{L_1^3} + \frac{L_2^3 - \frac{5}{2}L_2^2 + L_2}{L_1^4} \\ &\quad + \frac{L_2^4 - \frac{13}{3}L_2^3 + \frac{9}{2}L_2^2 - L_2}{L_1^5} + \mathcal{O}\left(\frac{L_2^5}{L_1^6}\right). \end{aligned} \quad (4.25)$$

Consequently,

$$u_0 = \lambda \left[\frac{1}{L_1} + \frac{L_2}{L_1^2} + \frac{L_2^2 - L_2}{L_1^3} + \cdots \right]. \quad (4.26)$$

Substituting (4.26) into (4.15) produces a purely polynomial-logarithmic expansion in λ^{-1} , L_1 , and L_2 . Keeping W_0 unexpanded is usually more compact and numerically superior.

4.6 Residual certification and Newton improvement

Let $a_M(y)$ denote the truncation in (4.15), and define the logarithmic residual

$$\rho_M(y) = \log \Gamma(a_M(y) + 1) - 1 - \log y. \quad (4.27)$$

Since the derivative with respect to a is $P(a)$, the first omitted error is certified by

$$a(y) - a_M(y) = -\frac{\rho_M(y)}{P(a_M(y))} + \mathcal{O}\left(\frac{\rho_M(y)^2 P'(a_M(y))}{P(a_M(y))^3}\right). \quad (4.28)$$

One Newton step,

$$a_M^N = a_M - \frac{\rho_M}{P(a_M)}, \quad (4.29)$$

therefore squares the local residual. This observation will recur for the full subfactorial inverse.

Computational checkpoint

The centered expansion is already effective at modest arguments. For $y = 10^6$, the Lambert seed errs by about 1.72×10^{-3} ; successively including p_1, p_2, p_3, p_4 reduces the absolute error to approximately 1.27×10^{-6} , 3.52×10^{-9} , 2.29×10^{-11} , and 2.85×10^{-13} , respectively. A detailed table appears in [Section 11](#).

5 A general reversion calculus for a rapid core plus a small tail

We now isolate the mechanism that will be applied to the subfactorial. The basic problem is

$$F(x) = \mathcal{A}(x) + \mathcal{R}(x), \quad y = F(x), \quad (5.1)$$

where $\mathcal{A}$ is eventually one-to-one and grows so rapidly that $\mathcal{R}(x)/\mathcal{A}(x) \rightarrow 0$. The inverse of $\mathcal{A}$ will be denoted by

$$a(y) = \mathcal{A}^{-1}(y). \quad (5.2)$$

5.1 Lagrange–Bürmann after straightening the core

The most efficient derivation begins by using the core itself as a coordinate. Put

$$u = \mathcal{A}(x), \quad x = a(u), \quad \rho(u) = \mathcal{R}(a(u)). \quad (5.3)$$

Then (5.1) becomes the near-identity equation

$$y = u + \rho(u). \quad (5.4)$$

Ordinary Lagrange–Bürmann inversion can now be used without attempting to reverse the rapid function directly.

Theorem 5.1 (Master reversion identity). *Introduce a bookkeeping parameter ε and let $x_\varepsilon(y)$ be the local inverse of*

$$F_\varepsilon(x) = \mathcal{A}(x) + \varepsilon \mathcal{R}(x) \quad (5.5)$$

that satisfies $x_0(y) = a(y)$. Whenever the local analytic inverse exists,

$$x_\varepsilon(y) = a(y) + \sum_{n=1}^{\infty} \frac{(-\varepsilon)^n}{n!} \frac{d^{n-1}}{dy^{n-1}} [a'(y) \mathcal{R}(a(y))^n]. \quad (5.6)$$

The identity is valid as a convergent Taylor series in ε near zero and, independently, as a formal identity. If the coefficients are asymptotically ordered when $y \rightarrow \infty$, setting $\varepsilon = 1$ gives the inverse transseries of F .

Proof. In the straightened coordinate u , the equation is $y = u + \varepsilon \rho(u)$. The Lagrange–Bürmann formula for an analytic test function H states

$$H(u(y)) = H(y) + \sum_{n=1}^{\infty} \frac{(-\varepsilon)^n}{n!} \frac{d^{n-1}}{dy^{n-1}} [H'(y) \rho(y)^n]. \quad (5.7)$$

Choose $H = a$. Since $x = a(u)$ and $\rho(y) = \mathcal{R}(a(y))$, this is exactly (5.6). $\square$

The first three terms, written without any structural assumptions, are

$$x(y) = a - a'R + \frac{1}{2} \frac{d}{dy} (a'R^2) - \frac{1}{6} \frac{d^2}{dy^2} (a'R^3) + \dots, \quad (5.8)$$

where $R = \mathcal{R}(a(y))$. The universal leading displacement is therefore

$$x - a \sim -\frac{\mathcal{R}(a)}{\mathcal{A}'(a)}. \quad (5.9)$$

This is the familiar “residual divided by slope” rule, now embedded in an all-orders formula.

5.2 The logarithmic derivative and outer valuation

Define the logarithmic derivative of the core by

$$\Lambda(x) = \frac{\mathcal{A}'(x)}{\mathcal{A}(x)}. \quad (5.10)$$

At $x = a(y)$,

$$a'(y) = \frac{1}{y \Lambda(a(y))}. \quad (5.11)$$

For any differentiable coefficient f and integer $r \geq 1$,

$$\frac{d}{dy} [y^{-r} f(a(y))] = y^{-r-1} \mathcal{D}_r f(a(y)), \quad (5.12)$$

where

$$\mathcal{D}_r f = \frac{f'}{\Lambda} - r f. \quad (5.13)$$

Every y -derivative therefore raises the outer y^{-1} valuation by one. This simple fact is the source of local finiteness in the resulting transseries.

Theorem 5.2 (Operator form of the inverse sectors). *Assume that $\mathcal{A}$ is eventually positive and increasing, and let $\Lambda = \mathcal{A}'/\mathcal{A}$. Then the formal inverse of $F = \mathcal{A} + \mathcal{R}$ around $a = \mathcal{A}^{-1}$ is*

$$F^{-1}(y) = a(y) + \sum_{n=1}^{\infty} \frac{X_n(a(y))}{y^n}, \quad (5.14)$$

with

$$X_n(x) = \frac{(-1)^n}{n!} (\mathcal{D}_{n-1} \circ \cdots \circ \mathcal{D}_1) \left(\frac{\mathcal{R}(x)^n}{\Lambda(x)} \right), \quad (5.15)$$

where an empty composition is the identity. In particular,

$$X_1 = -\frac{\mathcal{R}}{\Lambda}, \quad (5.16)$$

$$X_2 = \frac{1}{2} \left[\frac{1}{\Lambda} \left(\frac{\mathcal{R}^2}{\Lambda} \right)' - \frac{\mathcal{R}^2}{\Lambda} \right] \quad (5.17)$$

$$= \frac{\mathcal{R}\mathcal{R}'}{\Lambda^2} - \frac{\mathcal{A}''}{2\mathcal{A}} \frac{\mathcal{R}^2}{\Lambda^3}. \quad (5.18)$$

Proof. In the n th term of (5.6), use $a' = 1/(y\Lambda(a))$ to obtain

$$a'(y) \mathcal{R}(a(y))^n = y^{-1} \left(\frac{\mathcal{R}^n}{\Lambda} \right)_{x=a(y)}.$$

Applying (5.12) successively for $r = 1, 2, \dots, n-1$ proves (5.15). The first two coefficients follow directly. To pass from (5.17) to (5.18), note that $\Lambda' = \mathcal{A}''/\mathcal{A} - \Lambda^2$; the $\mathcal{R}^2/(2\Lambda)$ terms cancel. $\square$

5.3 An admissible asymptotic class and remainder theorem

The operator identity is formal without any estimates. The following conditions are convenient for a Poincaré interpretation. They are not minimal, but they cover the subfactorial and many gamma-type examples.

Definition 5.3 (Admissible rapid-core pair). A pair $(\mathcal{A}, \mathcal{R})$ on $[x_0, \infty)$ is called admissible if:

1. $\mathcal{A} > 0$ is analytic in a fixed complex neighborhood of the ray, is strictly increasing there, and $\mathcal{A}(x) \rightarrow \infty$;
2. $\Lambda = \mathcal{A}'/\mathcal{A} > 0$ and there is a positive scale $s(x)$ with $s(x) = o(\Lambda(x))$;
3. for every fixed $k \geq 0$,

$$\mathcal{R}^{(k)}(x) = \mathcal{O}(M(x)s(x)^k), \quad (\Lambda^{\pm 1})^{(k)}(x) = \mathcal{O}(\Lambda(x)^{\pm 1} s(x)^k),$$

for some positive $M(x)$;

4. $M(x)/\mathcal{A}(x) \rightarrow 0$.

Oscillatory factors $e^{i\omega x}$ with fixed ω are permitted by replacing s with $s + |\omega|$.

Theorem 5.4 (Fixed-sector asymptotic remainder). *For an admissible pair, suppose in addition that the local inverse branch of $\mathcal{A} + \varepsilon\mathcal{R}$ remains analytic for $|\varepsilon| \leq 1 + \eta$ for all sufficiently large y , with some fixed $\eta > 0$. Then, for every fixed N ,*

$$F^{-1}(y) = a(y) + \sum_{n=1}^N \frac{X_n(a(y))}{y^n} + \mathcal{O}_N\left(\frac{M(a(y))^{N+1}}{y^{N+1}\Lambda(a(y))}\right). \quad (5.19)$$

Moreover,

$$X_n(x) = \mathcal{O}_n\left(\frac{M(x)^n}{\Lambda(x)}\right). \quad (5.20)$$

Proof. Because $s/\Lambda \rightarrow 0$, the operator $\mathcal{D}_r = \Lambda^{-1}D_x - r$ maps a symbol of size M^n/Λ to one of the same size at fixed r, n . This proves (5.20) inductively. Analyticity in ε and Cauchy's estimate bound the Taylor remainder of (5.6) by the first neglected asymptotic scale, uniformly for sufficiently large y . The factor y^{-N-1} is explicit from the valuation-raising identity (5.12); the remaining coefficient has the size in (5.20). $\square$

Remark 5.5. The uniform analyticity hypothesis can be replaced by a direct Newton–Kantorovich or Rouché argument. In the subfactorial case the slope of the core is of order $y \log a$, whereas the tail and all of its fixed derivatives are only algebraic in a . A disk of radius $C/(ya \log a)$ around a is therefore more than sufficient.

5.4 Fourier closure and phase-sensitive operators

Suppose the tail has a finite or asymptotic Fourier decomposition

$$\mathcal{R}(x) = \sum_{\omega \in \Omega} e^{i\omega x} r_\omega(x), \quad (5.21)$$

where r_ω are slow symbols. Products in the n th sector create only frequencies in the n -fold additive sumset

$$n\Omega = \{\omega_1 + \cdots + \omega_n : \omega_j \in \Omega\}. \quad (5.22)$$

The coefficient field is closed because

$$\mathcal{D}_r(e^{i\omega x} f(x)) = e^{i\omega x} \left[\frac{f' + i\omega f}{\Lambda} - r f \right]. \quad (5.23)$$

Thus inversion never generates arbitrary new frequencies: it generates additive combinations of frequencies already present in the perturbation. For a single phase ω , sector n carries exactly the phase $e^{in\omega a(y)}$.

5.5 The frozen-coefficient logarithm

When D_x/Λ is asymptotically small on the coefficient field, each operator $\mathcal{D}_r$ is dominated by multiplication by $-r$. Therefore

$$X_n(x) \sim -\frac{1}{n} \frac{\mathcal{R}(x)^n}{\Lambda(x)}. \quad (5.24)$$

Summing these leading pieces gives the useful resummation

$$F^{-1}(y) \approx a(y) + \frac{1}{\Lambda(a(y))} \log \left(1 - \frac{\mathcal{R}(a(y))}{y} \right). \quad (5.25)$$

It should not be mistaken for an exact identity: derivatives of $\mathcal{R}$ and Λ supply the systematic corrections. Nevertheless, (5.25) captures the leading term in every outer sector at once.

5.6 Residual certification and Newton doubling

Let

$$G_N(y) = a(y) + \sum_{n=1}^N \frac{X_n(a(y))}{y^n} \quad (5.26)$$

be a truncated inverse. Its residual is

$$\mathcal{E}_N(y) = F(G_N(y)) - y. \quad (5.27)$$

Taylor's theorem gives

$$F^{-1}(y) - G_N(y) = -\frac{\mathcal{E}_N(y)}{F'(G_N(y))} + \mathcal{O}\left(\frac{F''(G_N(y))\mathcal{E}_N(y)^2}{F'(G_N(y))^3}\right). \quad (5.28)$$

Because $F'(a) \sim y\Lambda(a)$, a residual of outer order y^{-N} produces an inverse error of outer order y^{-N-1} . The Newton update

$$G_N^N = G_N - \frac{F(G_N) - y}{F'(G_N)} \quad (5.29)$$

then approximately doubles the resolved outer valuation. This is useful both numerically and as a symbolic consistency check.

5.7 A coefficient algorithm

Algorithm 5.6 (Rapid-core inverse sectors). Given symbolic descriptions of $\Lambda(x)$ and $\mathcal{R}(x)$ and a desired sector order N :

1. Set $X_0(x) = x$ only as a bookkeeping label.
2. For $n = 1, \dots, N$, initialize

$$H \leftarrow \frac{\mathcal{R}(x)^n}{\Lambda(x)}.$$

3. For $r = 1, \dots, n-1$, replace

$$H \leftarrow \frac{H'(x)}{\Lambda(x)} - rH.$$

4. Set $X_n(x) \leftarrow (-1)^n H/n!$.
5. Return

$$a(y) + \sum_{n=1}^N y^{-n} X_n(a(y)).$$

If $\mathcal{R}$ is itself represented by an asymptotic series, truncate its slow coefficient expansion only after the outer sector n has been fixed.

A scale-ordering trap

Expanding $a(y)$ into logarithms, expanding $\mathcal{R}(a)$ into inverse powers, and expanding in y^{-1} are three different truncations. They do not have the same valuation. The safe order is: first choose the outer sector order in y^{-1} ; next compute its coefficient exactly in a ; then expand that coefficient in the slow scale; finally replace a by a Lambert–Stirling approximation of sufficient accuracy. Reversing this order can discard terms that are larger than terms retained elsewhere.

6 Lateral inverse subfactorials: the complete Bell-sector expansion

We apply the general calculus to

$$\mathfrak{D}_\sigma(x) = \mathcal{A}(x) + \mathcal{R}_\sigma(x), \quad \mathcal{R}_\sigma(x) = e^{i\sigma\pi x} \mathcal{J}(x), \quad \Lambda(x) = P(x) = \psi(x+1). \quad (6.1)$$

The core inverse is $a = \mathfrak{a}$.

6.1 Existence and localization of the inverse branches

Theorem 6.1 (Distinguished large inverse branches). *There exists $y_0 > 0$ such that for every real $y > y_0$ and each $\sigma \in \{+1, -1\}$, the equation*

$$\mathfrak{D}_\sigma(z) = y \quad (6.2)$$

has exactly one solution $X_\sigma(y)$ in a disk

$$|z - \mathfrak{a}(y)| \leq \frac{C}{y\mathfrak{a}(y) \log \mathfrak{a}(y)}, \quad (6.3)$$

where C is independent of y . The solution is simple, depends analytically on y in a neighborhood of the positive ray, and satisfies

$$X_\sigma(y) - \mathfrak{a}(y) = -\frac{e^{i\sigma\pi\mathfrak{a}(y)} \mathcal{J}(\mathfrak{a}(y))}{yP(\mathfrak{a}(y))} + \mathcal{O}\left(\frac{1}{y^2 a^2 \log a}\right), \quad (6.4)$$

where $a = \mathfrak{a}(y)$ in the remainder.

Proof. Let $a = \mathfrak{a}(y)$ and write $z = a + \delta$. On a disk of radius $r = C/(ya \log a)$, Taylor's theorem and the standard asymptotics of the polygamma functions give

$$\mathcal{A}(a + \delta) - \mathcal{A}(a) = yP(a)\delta [1 + o(1)].$$

On the boundary its modulus is $C/a [1 + o(1)]$. Meanwhile $e^{i\sigma\pi z} \mathcal{J}(z) = \mathcal{O}(a^{-1})$ uniformly in the disk. Choosing C sufficiently large and then y sufficiently large makes the core difference dominate the tail on the boundary. Rouché's theorem shows that the perturbed equation has the same number of zeros as the core difference, namely one. Simplicity follows because the derivative is $yP(a)[1 + o(1)]$. One Taylor step gives (6.4); analytic dependence follows from the implicit-function theorem. $\square$

For real y , conjugation of (2.5) gives

$$X_-(y) = \overline{X_+(y)}. \quad (6.5)$$

Thus it is enough to compute the principal branch X_+ , but retaining σ keeps the branch symmetry visible.

6.2 The phase-adapted differential operators

For integers $r \geq 1$ and $n \geq 1$, define

$$\mathcal{L}_{r,n}^{(\sigma)} f = \frac{f'}{P} + \left(\frac{i\sigma n\pi}{P} - r \right) f. \quad (6.6)$$

Then

$$\frac{d}{dy} \left[y^{-r} e^{i\sigma n\pi a(y)} f(a(y)) \right] = y^{-r-1} e^{i\sigma n\pi a(y)} \mathcal{L}_{r,n}^{(\sigma)} f(a(y)). \quad (6.7)$$

This is (5.23) with $\omega = \sigma n\pi$ and $\Lambda = P$.

Theorem 6.2 (All-orders Bell-sector transseries). *For every fixed N and $y \rightarrow +\infty$,*

$$X_\sigma(y) = a(y) + \sum_{n=1}^N \frac{e^{i\sigma n\pi a(y)}}{y^n} C_{n,\sigma}(a(y)) + \mathcal{O}_N \left(\frac{1}{y^{N+1} a(y)^{N+1} \log a(y)} \right), \quad (6.8)$$

where $a = \mathfrak{a}$ and

$$C_{n,\sigma}(x) = \frac{(-1)^n}{n!} (\mathcal{L}_{n-1,n}^{(\sigma)} \circ \cdots \circ \mathcal{L}_{1,n}^{(\sigma)}) \left(\frac{\mathcal{J}(x)^n}{P(x)} \right). \quad (6.9)$$

At fixed n ,

$$C_{n,\sigma}(x) \sim -\frac{1}{n} \frac{\mathcal{J}(x)^n}{P(x)} \sim -\frac{1}{n x^n \log x}. \quad (6.10)$$

Proof. Apply Theorem 5.2 to $\mathcal{R}_\sigma = e^{i\sigma\pi x} \mathcal{J}(x)$ and move the phase $e^{i\sigma n\pi x}$ outside each phase-adapted operator using (6.7). Since $\mathcal{J}^{(k)}(x) = \mathcal{O}_k(x^{-k-1})$ and $P(x) \sim \log x$ while $P^{(k)}(x) = \mathcal{O}_k(x^{-k})$, the pair is admissible with $M(x) = x^{-1}$ and a fixed derivative scale. The remainder follows from Theorem 5.4 and the branch existence theorem.

For the leading coefficient, the derivative and phase terms divided by P are smaller than multiplication by $-r$. The composition contributes $(-1)^{n-1}(n-1)!$, which multiplied by $(-1)^n/n!$ gives $-1/n$. Finally use $\mathcal{J}(x) \sim x^{-1}$ and $P(x) \sim \log x$. $\square$

6.3 The first two sectors

The first coefficient is particularly simple:

$$C_{1,\sigma}(x) = -\frac{\mathcal{J}(x)}{P(x)}. \quad (6.11)$$

The second is

$$C_{2,\sigma}(x) = \frac{\mathcal{J}\mathcal{J}'}{P^2} + \frac{i\sigma\pi\mathcal{J}^2}{P^2} - \frac{\mathcal{J}^2 P'}{2P^3} - \frac{\mathcal{J}^2}{2P}. \quad (6.12)$$

Therefore

$$\begin{aligned} X_\sigma(y) = a - \frac{e^{i\sigma\pi a}}{y} \frac{\mathcal{J}(a)}{P(a)} \\ + \frac{e^{2i\sigma\pi a}}{y^2} \left[\frac{\mathcal{J}(a)\mathcal{J}'(a)}{P(a)^2} + \frac{i\sigma\pi\mathcal{J}(a)^2}{P(a)^2} - \frac{\mathcal{J}(a)^2 P'(a)}{2P(a)^3} - \frac{\mathcal{J}(a)^2}{2P(a)} \right] \\ + \mathcal{O} \left(\frac{1}{y^3 a^3 \log a} \right). \end{aligned} \quad (6.13)$$

This form is often more accurate than a much longer fully expanded formula, because $\mathcal{J}$, P , and a can all be evaluated directly.

6.4 Bell–logarithmic expansion inside the first sector

Let $\ell = \log x$. The digamma expansion

$$P(x) = \psi(x+1) \sim \ell + \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} - \frac{1}{252x^6} + \cdots \quad (6.14)$$

combined with (3.9) gives

$$\begin{aligned} \frac{\mathcal{J}(x)}{P(x)} &\sim \frac{1}{x\ell} - \frac{4\ell+1}{2x^2\ell^2} + \frac{60\ell^2+13\ell+3}{12x^3\ell^3} \\ &\quad - \frac{360\ell^3+64\ell^2+14\ell+3}{24x^4\ell^4} \\ &\quad + \frac{37440\ell^4+5694\ell^3+1025\ell^2+225\ell+45}{720x^5\ell^5} + \mathcal{O}\left(\frac{1}{x^6\ell}\right). \end{aligned} \quad (6.15)$$

The Bell numbers 1, 2, 5, 15, 52, ... are encoded in the numerators after division by the digamma series. No new transcendental constants occur in this first sector.

Substitution into (6.11) yields

$$\begin{aligned} -\frac{e^{i\sigma\pi a}}{y} \frac{\mathcal{J}(a)}{P(a)} &= -\frac{e^{i\sigma\pi a}}{y} \left[\frac{1}{a \log a} - \frac{4 \log a + 1}{2a^2(\log a)^2} \right. \\ &\quad \left. + \frac{60(\log a)^2 + 13 \log a + 3}{12a^3(\log a)^3} + \mathcal{O}\left(\frac{1}{a^4 \log a}\right) \right]. \end{aligned} \quad (6.16)$$

6.5 Expansion inside the second sector

Expanding (6.12) gives

$$\begin{aligned} C_{2,\sigma}(x) &= -\frac{\ell - 2i\sigma\pi}{2x^2\ell^2} \\ &\quad + \frac{8\ell^2 - 3\ell - 2 - 16i\sigma\pi\ell - 4i\sigma\pi}{4x^3\ell^3} \\ &\quad + \frac{-168\ell^3 + 119\ell^2 + 75\ell + 18 + 336i\sigma\pi\ell^2 + 100i\sigma\pi\ell + 18i\sigma\pi}{24x^4\ell^4} \\ &\quad + \mathcal{O}\left(\frac{1}{x^5\ell}\right). \end{aligned} \quad (6.17)$$

The appearance of π is entirely due to differentiating the oscillatory phase. The leading term $-1/(2x^2\ell)$ agrees with the universal law (6.10).

6.6 A compact nested transseries

Combining Theorems 4.1 and 6.2 gives a single all-orders description. Define

$$\widehat{a}(y) = -\frac{1}{2} + u_0 + \sum_{m \geq 1} \frac{p_m(w)}{u_0^{2m-1}}, \quad u_0 = \frac{\lambda}{W_0(\lambda/e)}, \quad \lambda = \log \frac{ey}{\sqrt{2\pi}}. \quad (6.18)$$

Then, with the understanding that each outer coefficient is expanded only to the slow order required,

$$\boxed{X_\sigma(y) \sim \widehat{a}(y) + \sum_{n \geq 1} \frac{e^{i\sigma n \pi \widehat{a}(y)}}{y^n} \widehat{C}_{n,\sigma}(\widehat{a}(y))}, \quad (6.19)$$

where $\widehat{C}_{n,\sigma}$ is obtained from (6.9) by substituting the Bell expansion of $\mathcal{J}$ and the Bernoulli expansion of P . If W_0 is further expanded by (4.25), every coefficient becomes a polynomial–logarithmic expression in $Y = \log y$, $\log Y$, and $\log \log Y$, multiplied by the outer monomial e^{-nY} and the chirped phase $\exp(i\sigma n\pi Y / \log Y + \dots)$.

6.7 Leading real and imaginary displacements

For the principal branch $\sigma = +1$, write $a = \mathfrak{a}(y)$ and $Q(a) = \mathcal{J}(a)/P(a) > 0$. The first sector gives

$$\Re X_+(y) = a - \frac{Q(a)}{y} \cos(\pi a) + \mathcal{O}\left(\frac{1}{y^2 a^2 \log a}\right), \quad (6.20)$$

$$\Im X_+(y) = -\frac{Q(a)}{y} \sin(\pi a) + \mathcal{O}\left(\frac{1}{y^2 a^2 \log a}\right). \quad (6.21)$$

The inverse branch therefore winds across the real axis whenever the core inverse passes an integer. At values $y = \mathcal{A}(n) = n!/e$, the leading shift is real because the phase is $(-1)^n$; at generic y it is complex.

6.8 Leading all-sector resummation

Applying (5.25) gives

$$X_\sigma(y) \approx a(y) + \frac{1}{P(a(y))} \log \left(1 - \frac{e^{i\sigma\pi a(y)} \mathcal{J}(a(y))}{y} \right). \quad (6.22)$$

Expanding the logarithm reproduces $-e^{i\sigma n\pi a} \mathcal{J}(a)^n / [ny^n P(a)]$, the leading term of every sector. Formula (6.22) is useful as a compact multi-sector initial guess for Newton iteration.

7 The real median inverse

The median interpolation

$$\mathfrak{D}_{\text{med}}(x) = \mathcal{A}(x) + \mathcal{J}(x) \cos(\pi x)$$

is eventually strictly increasing by Theorem 2.5; denote its large real inverse by $X_{\text{med}}(y)$. It can be treated either directly with Theorem 5.2 or as the average-frequency perturbation

$$\mathcal{J}(x) \cos(\pi x) = \frac{\mathcal{J}(x)}{2} (e^{i\pi x} + e^{-i\pi x}).$$

Unlike the average of the inverse branches, the inverse of the average is a nonlinear object; higher sectors contain mixed frequencies.

7.1 All-orders operator formula

Define

$$R_{\text{med}}(x) = \mathcal{J}(x) \cos(\pi x), \quad \mathcal{L}_r f = \frac{f'}{P} - r f. \quad (7.1)$$

Then

Theorem 7.1 (Median inverse transseries). *For every fixed N ,*

$$X_{\text{med}}(y) = a(y) + \sum_{n=1}^N \frac{C_n^{\text{med}}(a(y))}{y^n} + \mathcal{O}_N\left(\frac{1}{y^{N+1}a(y)^{N+1}\log a(y)}\right), \quad (7.2)$$

where

$$C_n^{\text{med}}(x) = \frac{(-1)^n}{n!} (\mathcal{L}_{n-1} \circ \dots \circ \mathcal{L}_1) \left(\frac{R_{\text{med}}(x)^n}{P(x)} \right). \quad (7.3)$$

The first two sectors are

$$C_1^{\text{med}}(x) = -\frac{\mathcal{J}(x) \cos(\pi x)}{P(x)}, \quad (7.4)$$

$$C_2^{\text{med}}(x) = \frac{1}{2} \left[\frac{1}{P(x)} \left(\frac{\mathcal{J}(x)^2 \cos^2(\pi x)}{P(x)} \right)' - \frac{\mathcal{J}(x)^2 \cos^2(\pi x)}{P(x)} \right]. \quad (7.5)$$

Proof. This is [Theorems 5.2 and 5.4](#) with $\mathcal{R} = R_{\text{med}}$ and $\Lambda = P$. The derivative of the cosine is bounded, so R_{med} belongs to the same admissible symbol class as the lateral tail. $\square$

The leading approximation is consequently

$$X_{\text{med}}(y) = a(y) - \frac{\mathcal{J}(a(y)) \cos(\pi a(y))}{yP(a(y))} + \mathcal{O}\left(\frac{1}{y^2 a(y)^2 \log a(y)}\right). \quad (7.6)$$

Its all-sector frozen-coefficient resummation is

$$X_{\text{med}}(y) \approx a(y) + \frac{1}{P(a(y))} \log \left(1 - \frac{\mathcal{J}(a(y)) \cos(\pi a(y))}{y} \right). \quad (7.7)$$

7.2 Frequency mixing

At sector n , the factor $\cos^n(\pi x)$ contains frequencies $(n - 2j)\pi$, $0 \leq j \leq n$. Thus

$$\cos^n \theta = 2^{-n} \sum_{j=0}^n \binom{n}{j} e^{i(n-2j)\theta}. \quad (7.8)$$

The phase-adapted operators preserve this finite set. In particular, even sectors contain a nonoscillatory component, while odd sectors do not. This is a useful illustration of the additive-frequency principle in [Equation \(5.22\)](#): taking a real median does not remove the oscillatory algebra; it symmetrizes it.

7.3 Comparison with the lateral branches

At first order,

$$X_{\text{med}}(y) - a(y) = \frac{(X_+(y) - a(y)) + (X_-(y) - a(y))}{2} + \mathcal{O}\left(\frac{1}{y^2 a^2 \log a}\right). \quad (7.9)$$

At second and higher orders this equality fails because inversion is nonlinear. The discrepancy can be computed directly from [\(6.12\)](#) and [\(7.5\)](#); it is generated by mixed products of the $+\pi$ and $-\pi$ frequencies.

8 The discrete generalized inverse and its threshold transseries

The integer sequence itself has the monotone tail $D_2 = 1, D_3 = 2, D_4 = 9, \dots$. Define its right-continuous generalized inverse for $y \geq 1$ by

$$N(y) = \min\{n \geq 2 : D_n \geq y\}. \quad (8.1)$$

This is a staircase, not a smooth function. Its asymptotic relation to the inverse gamma core is best encoded by the locations of its jumps.

8.1 Core-coordinate thresholds

Define

$$\alpha_n = \mathfrak{a}(D_n), \quad \text{equivalently} \quad \mathcal{A}(\alpha_n) = D_n. \quad (8.2)$$

Because $\mathfrak{a}$ is increasing,

$$N(y) = \min\{n \geq 2 : \mathfrak{a}(y) \leq \alpha_n\}. \quad (8.3)$$

The thresholds α_n are exponentially close to the integers.

Theorem 8.1 (Threshold displacement). *Let $s_n = (-1)^n$, $J_n = \mathcal{J}(n)$, and $P_n = \psi(n+1)$. Then*

$$\begin{aligned} \alpha_n - n &= \frac{s_n J_n}{\mathcal{A}'(n)} - \frac{\mathcal{A}''(n) J_n^2}{2\mathcal{A}'(n)^3} \\ &\quad + \frac{s_n (3\mathcal{A}''(n)^2 - \mathcal{A}'(n)\mathcal{A}'''(n)) J_n^3}{6\mathcal{A}'(n)^5} + \mathcal{O}\left(\frac{J_n^4}{\mathcal{A}(n)^4 P_n}\right). \end{aligned} \quad (8.4)$$

In particular,

$$\alpha_n = n + (-1)^n \frac{e\mathcal{J}(n)}{n!\psi(n+1)} + \mathcal{O}\left(\frac{1}{(n!)^2 n^2 \log n}\right). \quad (8.5)$$

Using the Bell expansion of $\mathcal{J}(n)$ yields a complete factorially small threshold transseries.

Proof. From (2.7), $D_n = \mathcal{A}(n) + s_n J_n$. Taylor-expand the local inverse of $\mathcal{A}$ at $\mathcal{A}(n)$. The standard inverse-function derivatives are

$$\begin{aligned} (\mathcal{A}^{-1})' &= \frac{1}{\mathcal{A}'}, \\ (\mathcal{A}^{-1})'' &= -\frac{\mathcal{A}''}{\mathcal{A}'^3}, \\ (\mathcal{A}^{-1})''' &= \frac{3\mathcal{A}''^2 - \mathcal{A}'\mathcal{A}'''}{\mathcal{A}'^5}. \end{aligned}$$

Substitution of the increment $s_n J_n$ gives (8.4). Since $\mathcal{A}'(n) = (n!/e)P_n$, $J_n = \mathcal{O}(n^{-1})$, and $\mathcal{A}''(n) = \mathcal{A}(n)(P_n^2 + P'_n)$, the first term is (8.5) and the second is of the stated factorial order. $\square$

The first term can be written entirely in elementary integer data because $P_n = H_n - \gamma$ and

$$\mathcal{J}(n) = (-1)^n \left(D_n - \frac{n!}{e} \right). \quad (8.6)$$

For asymptotic work, however, the Bell expansion is more revealing:

$$\alpha_n - n \sim \frac{(-1)^n e}{n!(H_n - \gamma)} \left(\frac{1}{n} - \frac{2}{n^2} + \frac{5}{n^3} - \frac{15}{n^4} + \dots \right) + \text{higher factorial sectors}. \quad (8.7)$$

8.2 Why the staircase has no ordinary smooth transseries

The difference $N(y) - \mathfrak{a}(y)$ remains bounded, but it does not approach a limit as y moves continuously through successive intervals $(D_{n-1}, D_n]$. In fact it traverses essentially a full unit interval between jumps. Hence there is no expansion

$$N(y) = \mathfrak{a}(y) + c_0 + o(1) \quad (8.8)$$

uniformly for all real $y \rightarrow \infty$ with a fixed constant c_0 . The correct decomposition is

$$N(y) = \lceil \mathfrak{a}(y) - \vartheta(y) \rceil, \quad (8.9)$$

where the threshold correction ϑ is supported through the tiny shifts $\alpha_n - n$. Formula (8.3), rather than a smooth Poincaré series, is the exact asymptotic normal form.

Remark 8.2 (Away from thresholds). If $\mathfrak{a}(y)$ stays a distance much larger than $1/(n!n \log n)$ from every integer n , then the exponentially small shifts in (8.5) do not affect the result and $N(y) = \lceil \mathfrak{a}(y) \rceil$. Only in exponentially narrow windows near integer core coordinates does the parity-dependent derangement correction change the step location.

9 Endpoint-Mellin tails and coefficient sequences

The Bell numbers arise from one particular endpoint kernel, but the mechanism is much broader. Let φ be analytic in a neighborhood of $t = 1$ and define

$$J_\varphi(x) = \int_0^1 t^x \varphi(t) dt. \quad (9.1)$$

With $t = e^{-s}$,

$$J_\varphi(x) = \int_0^\infty e^{-(x+1)s} g(s) ds, \quad g(s) = \varphi(e^{-s}). \quad (9.2)$$

Theorem 9.1 (Endpoint-jet expansion). *Suppose g is analytic at $s = 0$ and has sufficient growth control for Watson's lemma. Then*

$$J_\varphi(x) \sim \sum_{r=0}^{\infty} \frac{g^{(r)}(0)}{(x+1)^{r+1}}. \quad (9.3)$$

Equivalently, after re-expansion in powers of x^{-1} ,

$$J_\varphi(x) \sim \sum_{k=1}^{\infty} \frac{\mu_k}{x^k}, \quad \mu_k = \sum_{r=0}^{k-1} (-1)^{k-r-1} \binom{k-1}{r} g^{(r)}(0). \quad (9.4)$$

Proof. Insert the Taylor series $g(s) = \sum_{r \geq 0} g^{(r)}(0) s^r / r!$ into (9.2) and integrate termwise asymptotically:

$$\int_0^\infty e^{-(x+1)s} s^r ds = \frac{r!}{(x+1)^{r+1}}.$$

This proves (9.3). Expanding $(x+1)^{-r-1} = x^{-r-1}(1+x^{-1})^{-r-1}$ and collecting the coefficient of x^{-k} gives (9.4). $\square$

For the subfactorial,

$$\varphi(t) = e^{t-1}, \quad g(s) = e^{e^{-s}-1}, \quad g^{(r)}(0) = (-1)^r B_r. \quad (9.5)$$

Thus (9.3) becomes the shifted Bell expansion (3.17). Other endpoint kernels generate other coefficient sequences. For example:

- $\varphi(t) = 1$ gives $J_\varphi(x) = 1/(x+1)$ and a geometric sequence;
- $\varphi(t) = t^\beta$ gives $1/(x+\beta+1)$;
- $\varphi(t) = \exp(\alpha t)$ gives coefficients governed by Touchard polynomials $T_r(\alpha)$;
- products or compositions in φ produce complete Bell polynomials in its endpoint derivatives.

9.1 Inverse transseries for gamma-type endpoint perturbations

Consider

$$F(x) = \mathcal{A}(x) + \sum_{q=1}^Q e^{i\omega_q x} J_{\varphi_q}(x), \quad (9.6)$$

where $\mathcal{A}$ is a rapid core and the frequencies are fixed. Combining [Theorems 5.2](#) and [9.1](#) yields the nested expansion

$$F^{-1}(y) \sim a(y) + \sum_{n \geq 1} \frac{1}{y^n} \sum_{\omega \in n\Omega} e^{i\omega a(y)} C_{n,\omega}(a(y)), \quad (9.7)$$

where each $C_{n,\omega}$ has an inverse-power expansion determined by the endpoint jets of the φ_q and by the logarithmic derivative of the core. The subfactorial is the one-frequency case $Q = 1$, $\omega_1 = \sigma\pi$, and $\varphi_1(t) = e^{t-1}$.

10 Further structure of the rapid-core calculus

10.1 Functoriality under exact changes of core coordinate

The derivation of [Theorem 5.1](#) shows that the entire calculus is functorial under the coordinate change $u = \mathcal{A}(x)$. Algebraically, the inverse problem factors as

$$(\mathcal{A} + \mathcal{R})^{-1} = a \circ (\text{id} + \mathcal{R} \circ a)^{-1}. \quad (10.1)$$

This factorization separates two tasks:

1. reverse the near-identity map $u \mapsto u + \rho(u)$;
2. transport the result back through the exact core inverse a .

The coefficient formula [\(5.6\)](#) is precisely the transport of ordinary near-identity reversion through this factorization. It therefore inherits associativity and composition laws from the usual group of formal diffeomorphisms tangent to the identity.

10.2 Bell polynomials inside Bell sectors

There are two unrelated but interacting appearances of “Bell” in this problem.

1. The numerical Bell numbers B_k arise from the endpoint kernel and constitute the coefficients of $\mathcal{J}(x)$.
2. Faà di Bruno and repeated differentiation in [\(6.9\)](#) organize products of derivatives through partial Bell polynomials.

To make the second appearance explicit, write

$$\rho(y) = \mathcal{R}(a(y)). \quad (10.2)$$

Then

$$\frac{d^{n-1}}{dy^{n-1}} (a'(y)\rho(y)^n) = \sum_{j=0}^{n-1} \binom{n-1}{j} a^{(j+1)}(y) \frac{d^{n-1-j}}{dy^{n-1-j}} \rho(y)^n. \quad (10.3)$$

The last derivative can be expressed through exponential partial Bell polynomials in $\rho', \rho'', \dots$. Thus the endpoint Bell numbers enter as data, while combinatorial Bell polynomials enter as the universal chain rule of inversion.

10.3 Multiple algebraic tails

Suppose

$$\mathcal{R}(x) \sim \sum_{q \geq q_0} r_q (\log x) x^{-q} \quad (10.4)$$

with polynomial or rational logarithmic coefficients. At outer sector n , $\mathcal{R}(x)^n$ begins at x^{-nq_0} . The operator $\Lambda^{-1}D_x - r$ preserves the leading algebraic order when Λ grows logarithmically or more rapidly. Consequently,

$$X_n(x) = \mathcal{O}\left(\frac{x^{-nq_0}}{\Lambda(x)}\right). \quad (10.5)$$

This gives the support of the inverse transseries before any coefficient is computed.

10.4 Cores of exponential form

If

$$\mathcal{A}(x) = \exp(\Phi(x)), \quad (10.6)$$

then $\Lambda = \Phi'$, and the master coefficients simplify to

$$X_n(x) = \frac{(-1)^n}{n!} \left(\frac{D_x}{\Phi'} - (n-1) \right) \cdots \left(\frac{D_x}{\Phi'} - 1 \right) \left(\frac{\mathcal{R}(x)^n}{\Phi'(x)} \right). \quad (10.7)$$

Gamma-type functions fit this form with $\Phi(x) = \log \Gamma(x+1) - 1$. Barnes-type and generalized hyperfactorial cores also fit it after taking logarithms; only the preliminary inversion of Φ changes. The perturbative sector calculus remains identical.

10.5 A universality principle

Proposition 10.1 (Universal leading sector law). *Let $(\mathcal{A}, \mathcal{R})$ be admissible and suppose, for every fixed n ,*

$$D_x \log(\mathcal{R}^n / \Lambda) = o(\Lambda).$$

Then

$$F^{-1}(y) - a(y) \sim -\frac{1}{\Lambda(a(y))} \sum_{n \geq 1} \frac{1}{n} \left(\frac{\mathcal{R}(a(y))}{y} \right)^n \quad (10.8)$$

in the sense of matching the leading coefficient in each fixed outer sector. Equivalently, the sector leaders are generated by the logarithm in (5.25).

Proof. Under the stated hypothesis, every D_x/Λ contribution in (5.15) is lower order. The remaining composition is $(-1)^{n-1}(n-1)!$, proving (5.24). Summation over n is the Taylor series of $\log(1-z)$. $\square$

10.6 When the logarithmic resummation fails

The hypothesis of [Theorem 10.1](#) is essential. It fails if the tail contains phases whose frequencies grow as fast as Λ , boundary layers whose derivatives amplify the coefficient, or rapidly varying singularities near the inverse point. In such cases D_x/Λ and multiplication by $-r$ contribute at the same order, and the leading all-sector sum is no longer logarithmic. The exact operator formula (5.15) remains valid and should be used instead.

11 Numerical validation

The numerical tests in this section use high-precision arithmetic. The core inverse $a = \mathfrak{a}(y)$ is obtained by Newton iteration applied to

$$\log \Gamma(a + 1) - 1 - \log y = 0, \quad (11.1)$$

starting from the Lambert seed $u_0 - 1/2$. The principal lateral inverse is then obtained from Newton iteration on

$$\frac{\Gamma(z + 1)}{e} + e^{i\pi z} \mathcal{J}(z) - y = 0, \quad (11.2)$$

starting from the first-sector approximation. The endpoint integral is evaluated directly; the partial-fraction series (3.1) provides an independent check.

11.1 Accuracy of the centered inverse-gamma expansion

Let $a^{[M]}$ be the truncation containing $p_1, \dots, p_M$, with $a^{[0]} = u_0 - 1/2$. [Table 1](#) reports $|a^{[M]} - a|$.

Table 1: Absolute errors in the centered inverse-gamma expansion.

y	exact $a = \mathfrak{a}(y)$	$M = 0$	$M = 1$	$M = 2$	$M = 3$	$M = 4$
10^3	6.69048431391006103	2.9344×10^{-3}	4.7978×10^{-6}	2.7535×10^{-8}	3.6454×10^{-10}	9.2245×10^{-12}
10^6	9.87684651274886097	1.7154×10^{-3}	1.2715×10^{-6}	3.5241×10^{-9}	2.2938×10^{-11}	2.8457×10^{-13}
10^{10}	13.5617402632380633	1.1206×10^{-3}	4.3640×10^{-7}	6.6159×10^{-10}	2.3767×10^{-12}	1.6235×10^{-14}
10^{50}	41.5613050474933070	2.6492×10^{-4}	1.0626×10^{-8}	1.8221×10^{-12}	7.5138×10^{-16}	5.8501×10^{-19}

The consistent reduction by several orders of magnitude confirms both the half-shifted normal form and the rational functions (4.16)–(4.19). For still larger M , the Stirling series is asymptotic rather than convergent; the optimal truncation and any desired hyperasymptotic continuation should be handled through the exact residual (4.27).

11.2 Accuracy of the Bell sectors

Let

$$X_+^{[0]}(y) = a(y), \quad (11.3)$$

$$X_+^{[1]}(y) = a(y) + \frac{e^{i\pi a(y)}}{y} C_{1,+}(a(y)), \quad (11.4)$$

$$X_+^{[2]}(y) = X_+^{[1]}(y) + \frac{e^{2i\pi a(y)}}{y^2} C_{2,+}(a(y)). \quad (11.5)$$

[Table 2](#) compares these approximations with a direct high-precision root of (11.2).

Table 2: Principal lateral inverse: exact displacement and sector errors.

y	$X_+(y) - a(y)$	$ X_+^{[0]} - X_+ $	$ X_+^{[1]} - X_+ $	$ X_+^{[2]} - X_+ $
10^3	$3.32599835278648 \times 10^{-5} -$ $4.87639893172798 \times 10^{-5}i$	5.9027×10^{-5}	1.1643×10^{-8}	3.4746×10^{-12}
10^6	$-3.35445523390485 \times 10^{-8} +$ $1.36671789756972 \times 10^{-8}i$	3.6222×10^{-8}	4.4488×10^{-15}	8.2808×10^{-22}
10^{10}	$-4.70356185958560 \times 10^{-13} +$ $2.39449487665544 \times 10^{-12}i$	2.4403×10^{-12}	2.0481×10^{-23}	2.6085×10^{-34}

The ratios are consistent with one extra factor of approximately $1/(ya)$ for each added sector. At $y = 10^{10}$, two Bell sectors improve the core approximation from about twelve decimal orders in the displacement to roughly thirty-four decimal orders in absolute error.

11.3 Residual-based verification

A coefficient formula can be checked without knowing the exact inverse. Substitute a truncation $G_N(y)$ into the defining equation and expand

$$\mathfrak{D}_\sigma(G_N(y)) - y. \quad (11.6)$$

The coefficients through outer order y^{-N+1} must vanish, and the first nonzero residual must be of order $y^{-N}a^{-N-1}$ up to slow logarithmic factors. Division by $\mathfrak{D}'_\sigma(G_N) \sim y \log a$ then gives the inverse error predicted in (6.8). This residual test was used independently of the operator derivation for the first two sectors.

12 Implementation details and symbolic workflow

12.1 Stable numerical evaluation

For large y , direct evaluation of $\Gamma(a+1)/e - y$ can overflow even when the residual is small. The core equation should be solved in logarithmic form, as in (11.1). A robust procedure is:

Algorithm 12.1 (Numerical principal inverse subfactorial). Given $y > 0$ sufficiently large and a working precision of p digits:

1. Compute $\lambda = \log(ey/\sqrt{2\pi})$, $w = W_0(\lambda/e)$, and $a_0 = \lambda/w - 1/2$.
2. Optionally add p_1/u_0 , p_2/u_0^3 , and further centered corrections.
3. Apply Newton iteration to

$$h(a) = \log \Gamma(a+1) - 1 - \log y, \quad h'(a) = \psi(a+1),$$

to obtain the exact core inverse a at the desired precision.

4. Evaluate $\mathcal{J}(a)$ either by the integral $\int_0^1 e^{t-1} t^a dt$ or by (3.1). For large positive a , a short optimally truncated Bell expansion with its signed remainder bound is also effective.
5. Form the first-sector guess

$$z_0 = a - \frac{e^{i\pi a} \mathcal{J}(a)}{y\psi(a+1)}.$$

6. Apply complex Newton iteration to

$$H(z) = \mathfrak{D}_+(z) - y,$$

using

$$H'(z) = \frac{\Gamma(z+1)}{e} \psi(z+1) + e^{i\pi z} (\mathcal{J}'(z) + i\pi \mathcal{J}(z)).$$

Because the starting error is already of second Bell-sector order, very few Newton steps are normally required.

12.2 Symbolic construction of the slow coefficients

The two recurrences in this article are triangular in different senses.

Core recurrence.

Equation (4.23) determines $p_m(w)$ from $p_1, \dots, p_{m-1}$ by coefficient extraction in the dummy variable $r = u_0^{-1}$.

Outer-sector recurrence.

Equation (6.9) determines the complete n th Bell sector by $n - 1$ applications of a first-order differential operator. It does not require solving a new implicit equation.

A computer algebra implementation should represent the slow coefficient field as a truncated differential algebra generated by

$$x^{-1}, \quad (\log x)^{-1}, \quad P(x), \quad P'(x), \dots, \quad \mathcal{J}(x), \quad \mathcal{J}'(x), \dots, \quad (12.1)$$

and only afterward substitute the Bell and Bernoulli expansions. This prevents expression swell and preserves exact identities such as (6.12).

12.3 Pseudocode

```
# Core coefficients p_m(w)
Delta = 0
for m in 1..M:
    E = w*Delta + (1/r + Delta)*log(1 + r*Delta)
    E += sum(c[j]*r^(2*j-1)*(1+r*Delta)^(-(2*j-1)), j=1..m)
    p[m] = -coeff(E, r, 2*m-1)/(1+w)
    Delta += p[m]*r^(2*m-1)

# Bell sectors for the lateral inverse
for n in 1..N:
    H = J(x)^n / P(x)
    for r in 1..n-1:
        H = diff(H,x)/P(x) + (I*sigma*n*pi/P(x) - r)*H
    C[n] = (-1)^n * H / factorial(n)

X = a(y) + sum(exp(I*sigma*n*pi*a(y))*C[n](a(y))/y^n, n=1..N)
```

12.4 Precision management

The correction $X_+(y) - a(y)$ is of order $1/(ya \log a)$ and may be far smaller than the magnitude of a . If that correction is required to p decimal digits relative to itself, the core a must be computed with roughly

$$p + \log_{10} y + \log_{10} a + \log_{10} \log a \quad (12.2)$$

absolute guard digits beyond the decimal point. Alternatively, solve directly for the displacement $\delta = z - a$ using the normalized equation

$$\frac{\mathcal{A}(a + \delta)}{\mathcal{A}(a)} - 1 + \frac{e^{i\pi(a+\delta)} \mathcal{J}(a + \delta)}{y} = 0, \quad (12.3)$$

which avoids subtracting two nearly equal large numbers.

13 Interpretation of the transseries hierarchy

13.1 Outer and inner asymptotics

The exact sector formula (6.8) is an asymptotic expansion in the outer parameter $1/y$ with coefficients depending on the slow variable $a(y)$. Each coefficient has its own expansion in $1/a$ and $1/\log a$. Symbolically,

$$X_\sigma(y) - a(y) \sim \sum_{n \geq 1} y^{-n} e^{i\sigma n \pi a(y)} \sum_{p \geq n} \sum_{q \geq 1} c_{n,p,q}^{(\sigma)} a(y)^{-p} (\log a(y))^{-q}. \quad (13.1)$$

The lower bound $p \geq n$ follows because $\mathcal{J}^n$ begins with a^{-n} . At fixed n and p , only finitely many logarithmic structures occur at a given truncation order.

13.2 Ordering relative to the core expansion

The algebraic Stirling correction $u_0^{-(2m-1)}$ is much larger than any fixed Bell sector y^{-n} as $y \rightarrow \infty$. Hence every finite number of Stirling terms precedes the first Bell sector. The centered Stirling series is divergent at sufficiently high order, so a fully hyperasymptotic ordering can contain exponentially improved gamma-core remainders. The exact-core form (6.8) deliberately avoids asserting an artificial global order between those hyperasymptotic core contributions and the Bell sectors. It says instead:

Resolve the gamma core to whatever accuracy is required, then add the subfactorial-specific sectors as an independent perturbative hierarchy.

This modular viewpoint is particularly important at astronomically large arguments, where the exponentially improved remainder of a deeply truncated Stirling expansion may cross the scale y^{-1} .

13.3 Phase as a chirp in logarithmic time

With $Y = \log y$,

$$a(y) \sim \frac{Y}{W_0(Y/e)}. \quad (13.2)$$

The principal phase is therefore

$$e^{i\pi a(y)} \sim \exp\left(i\pi \frac{Y}{W_0(Y/e)}\right). \quad (13.3)$$

Its instantaneous frequency with respect to Y is

$$\frac{d}{dY}(\pi a(e^Y)) = \frac{\pi}{P(a(e^Y))} \sim \frac{\pi}{\log a(e^Y)}. \quad (13.4)$$

Thus the phase oscillates indefinitely but ever more slowly in logarithmic time Y . With respect to y itself, its derivative is smaller by another factor $1/y$.

13.4 Stokes and branch data

The lateral sign σ is not an arbitrary decoration: it records how the negative argument in the incomplete gamma function is approached. It is therefore part of the analytic data of the problem. The two transseries are complex conjugates on the positive real y axis, and their median gives a real interpolation only after the forward functions are averaged. Choosing one continuation and silently taking the real part of its inverse would not produce the inverse of any canonically specified forward function.

14 Extensions, conjectures, and research directions

14.1 General winding branches

For the branch (2.6) with $\omega_k = (2k + 1)\pi$, all formulas remain valid after replacing $\sigma\pi$ by ω_k . In particular,

$$C_{n,k}(x) = \frac{(-1)^n}{n!} \prod_{r=n-1}^1 \left[\frac{D_x}{P} + \frac{i n \omega_k}{P} - r \right] \left(\frac{\mathcal{J}(x)^n}{P(x)} \right), \quad (14.1)$$

where the product is composition from right to left. At fixed k , the same error estimate holds. Uniformity as $|k| \rightarrow \infty$ is a separate multiple-scale problem because the phase derivative $|\omega_k|/P(x)$ need no longer be small.

14.2 Transseries for parameter-dependent derangements

Many generalized derangement numbers have exponential generating functions of the form

$$\frac{\exp(-\alpha z)}{(1 - z)^\beta} \quad (14.2)$$

or finite products of such factors. Their dominant continuation is often a gamma-type core multiplied by powers, while the truncation error is an endpoint-Mellin integral. The two-stage method developed here applies: first invert the dominant logarithmic phase, then use (5.15) for the endpoint correction. Touchard, Bell, Stirling, or generalized Bell polynomials will appear according to the endpoint kernel.

14.3 Large-order behavior of the centered coefficients

The coefficients $p_m(w)$ inherit the factorial divergence of the Stirling series. The initial examples suggest the structured form

$$p_m(w) = \frac{(-1)^{m+1} P_m(w)}{d_m (1 + w)^{2m-1}}, \quad \deg P_m = 2m - 2, \quad (14.3)$$

with P_m having positive coefficients for the computed orders.

Conjecture 14.1 (Positivity pattern for centered inverse-gamma coefficients). *After clearing the minimal positive integer denominator, the polynomial $(-1)^{m+1}(1+w)^{2m-1}p_m(w)$ has nonnegative coefficients for every $m \geq 1$ and is strictly positive for $w \geq 0$.*

A proof would likely require a sign-preserving reformulation of the triangular recurrence (4.23); the alternating Bernoulli input makes coefficientwise positivity far from automatic.

14.4 Large-order behavior of Bell sectors

At fixed x , the Lagrange series in the bookkeeping parameter has a radius determined by the nearest critical point of $\mathcal{A}(x) + \varepsilon\mathcal{R}(x)$. As $n \rightarrow \infty$ with y fixed, the coefficients $C_{n,\sigma}$ will therefore show a different large-order law from the fixed- n , $y \rightarrow \infty$ asymptotics developed here. Locating the nearest coalescing inverse branches would determine that law and the optimal outer-sector truncation.

14.5 Resurgent completion

The centered Stirling series for the gamma core and the outer Bell-sector series have different sources of divergence. A full resurgent analysis would combine:

1. Borel singularities of the log-gamma Stirling series;
2. singularities in the Lagrange parameter caused by critical points of the perturbed inverse problem;
3. the incomplete-gamma branch data encoded by σ ;
4. possible scale crossings between exponentially improved core terms and y^{-n} Bell sectors.

The exact-core normal form (6.8) is a convenient starting point because it cleanly separates the first item from the remaining three.

14.6 Formalization prospects

The core pieces are well suited to formal proof assistants:

- the exact series (3.1) and remainder identity (3.7) reduce to dominated convergence, a finite geometric identity, and Dobiński's formula;
- the master inversion formula (5.6) can be developed in a filtered differential algebra of formal series;
- the operator recurrence (5.15) is finite at every requested sector order;
- residual certification (5.28) separates formal coefficient cancellation from analytic error bounds.

For the ProveIt corpus, a natural first milestone is a formal theorem for the Bell remainder bound and a purely algebraic implementation of the valuation-raising operators. The analytic existence theorem can then be added using established gamma and polygamma estimates.

15 Conclusion

The asymptotic inverse of the subfactorial is not obtained by merely replacing $n!$ with a continuous factorial and applying one Lambert W . The complete structure is

$$\boxed{\text{inverse gamma core} + \text{Lambert-Stirling corrections} + \text{oscillatory Bell sectors in } y^{-1}.} \quad (15.1)$$

The exact decomposition

$$\mathfrak{D}_\sigma(z) = \Gamma(z+1)/e + e^{i\sigma\pi z} \mathcal{J}(z)$$

turns the problem into a rapid-core perturbation. The endpoint tail $\mathcal{J}$ has Bell coefficients with an explicit signed remainder, and Lagrange-Bürmann inversion transports those coefficients into a locally finite hierarchy of inverse sectors. The distinguished principal branch is complex, its lower lateral companion is its conjugate, and a real inverse requires a separately specified real interpolation. The discrete inverse is a staircase whose asymptotic information is concentrated in factorially small shifts of its jump thresholds.

The reusable result is the operator calculus

$$X_n(x) = \frac{(-1)^n}{n!} \left(\frac{D_x}{\Lambda} - (n-1) \right) \cdots \left(\frac{D_x}{\Lambda} - 1 \right) \left(\frac{\mathcal{R}(x)^n}{\Lambda(x)} \right),$$

with phase-adapted variants for oscillatory tails. It applies whenever a function consists of a rapidly varying invertible core plus a tail small in relative size. The subfactorial supplies a particularly clean model because its tail is both exactly representable and combinatorially structured.

A Initial Bell data and endpoint coefficients

The first Bell numbers are

k	0	1	2	3	4	5	6	7	8	9	10	11
B_k	1	1	2	5	15	52	203	877	4140	21147	115975	678570

Hence

$$\begin{aligned} \mathcal{J}(x) \sim & x^{-1} - 2x^{-2} + 5x^{-3} - 15x^{-4} + 52x^{-5} - 203x^{-6} \\ & + 877x^{-7} - 4140x^{-8} + 21147x^{-9} - 115975x^{-10} + 678570x^{-11} - \dots \end{aligned} \quad (\text{A.1})$$

The shifted form is

$$\begin{aligned} \mathcal{J}(x) \sim & (x+1)^{-1} - (x+1)^{-2} + 2(x+1)^{-3} - 5(x+1)^{-4} \\ & + 15(x+1)^{-5} - 52(x+1)^{-6} + 203(x+1)^{-7} - \dots \end{aligned} \quad (\text{A.2})$$

B Derivation of the second inverse sector

From (6.9),

$$C_{2,\sigma} = \frac{1}{2} \mathcal{L}_{1,2}^{(\sigma)} \left(\frac{\mathcal{J}^2}{P} \right).$$

By definition,

$$\begin{aligned}\mathcal{L}_{1,2}^{(\sigma)}\left(\frac{\mathcal{J}^2}{P}\right) &= \frac{1}{P}\left(\frac{\mathcal{J}^2}{P}\right)' + \left(\frac{2i\sigma\pi}{P} - 1\right)\frac{\mathcal{J}^2}{P} \\ &= \frac{2\mathcal{J}\mathcal{J}'}{P^2} - \frac{\mathcal{J}^2 P'}{P^3} + \frac{2i\sigma\pi\mathcal{J}^2}{P^2} - \frac{\mathcal{J}^2}{P}.\end{aligned}$$

Division by two gives (6.12). This calculation also shows how phase differentiation and slow differentiation enter at different powers of $1/P$.

C First terms of the third sector

The exact compact formula is

$$C_{3,\sigma}(x) = -\frac{1}{6}\mathcal{L}_{2,3}^{(\sigma)}\mathcal{L}_{1,3}^{(\sigma)}\left(\frac{\mathcal{J}(x)^3}{P(x)}\right). \quad (\text{C.1})$$

Without expanding the full rational expression, its leading slow terms are

$$C_{3,\sigma}(x) = -\frac{1}{3x^3 \log x} + \mathcal{O}\left(\frac{1}{x^3(\log x)^2} + \frac{1}{x^4 \log x}\right). \quad (\text{C.2})$$

Thus the first three outer sectors begin

$$X_\sigma(y) - a(y) = -\frac{e^{i\sigma\pi a}}{ya \log a} - \frac{e^{2i\sigma\pi a}}{2y^2 a^2 \log a} - \frac{e^{3i\sigma\pi a}}{3y^3 a^3 \log a} + \text{lower slow orders}. \quad (\text{C.3})$$

These are precisely the first three terms in the logarithmic resummation (6.22) after replacing $\mathcal{J}(a)$ by $1/a$ and $P(a)$ by $\log a$.

D A direct formal proof of local finiteness

Let $\mathcal{K}$ be a differential coefficient field in the variable x containing $\Lambda^{\pm 1}$ and $\mathcal{R}$, and consider formal sums

$$\mathcal{K}((y^{-1}))_{\geq 0} = \left\{ \sum_{n \geq 0} y^{-n} f_n(a(y)) : f_n \in \mathcal{K} \right\}. \quad (\text{D.1})$$

Define the outer valuation of a nonzero series by

$$\nu\left(\sum_{n \geq n_0} y^{-n} f_n(a(y))\right) = n_0. \quad (\text{D.2})$$

Equation (5.12) implies

$$\nu(D_y S) \geq \nu(S) + 1. \quad (\text{D.3})$$

The n th Lagrange term in (5.6) starts with one explicit factor y^{-1} and contains $n - 1$ derivatives. It therefore has valuation at least n . Consequently, the coefficient of y^{-N} receives contributions only from Lagrange terms with $n \leq N$. This is local finiteness, and it justifies all formal coefficient extractions without any analytic convergence assumption.

E Notation crosswalk

Symbol	Meaning
$D_n = !n$	integer derangement number or subfactorial
$\mathfrak{D}_\sigma(z)$	lateral incomplete-gamma continuation, $\arg(-1) = \sigma\pi$
$\mathfrak{D}_{\text{med}}(z)$	real median interpolation $(\mathfrak{D}_+ + \mathfrak{D}_-)/2$
$\mathcal{A}(x)$	gamma core $\Gamma(x+1)/e$ in the subfactorial application; a general rapid core elsewhere
$\mathcal{J}(x)$	endpoint tail $e^{-1} \int_0^1 e^{tx} dt$
$\mathfrak{a}(y)$	large real inverse of $\mathcal{A}$, equal to $\Gamma_0^{-1}(ey) - 1$
$P(x)$	$\psi(x+1)$, the logarithmic derivative of the gamma core
$\Lambda(x)$	general logarithmic derivative $\mathcal{A}'(x)/\mathcal{A}(x)$
$X_\sigma(y)$	distinguished large inverse branch of $\mathfrak{D}_\sigma$
$C_{n,\sigma}(x)$	slow coefficient of the n th lateral Bell sector
λ	$\log(ey/\sqrt{2\pi})$
w	$W_0(\lambda/e)$
u_0	Lambert seed λ/w
$p_m(w)$	centered inverse-gamma Stirling coefficient
$N(y)$	discrete generalized inverse $\min\{n : D_n \geq y\}$
α_n	core-coordinate threshold $\mathfrak{a}(D_n)$

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